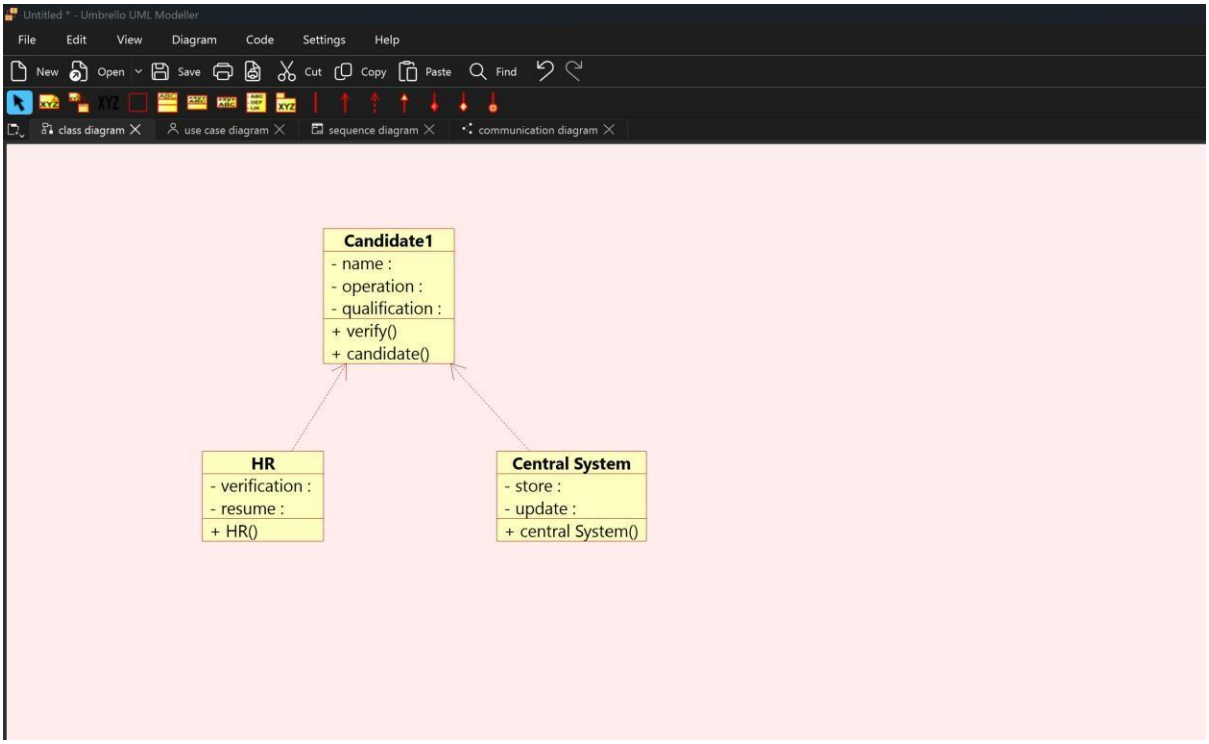
Ex No: 10 – RECRUITMENT SYSTEM

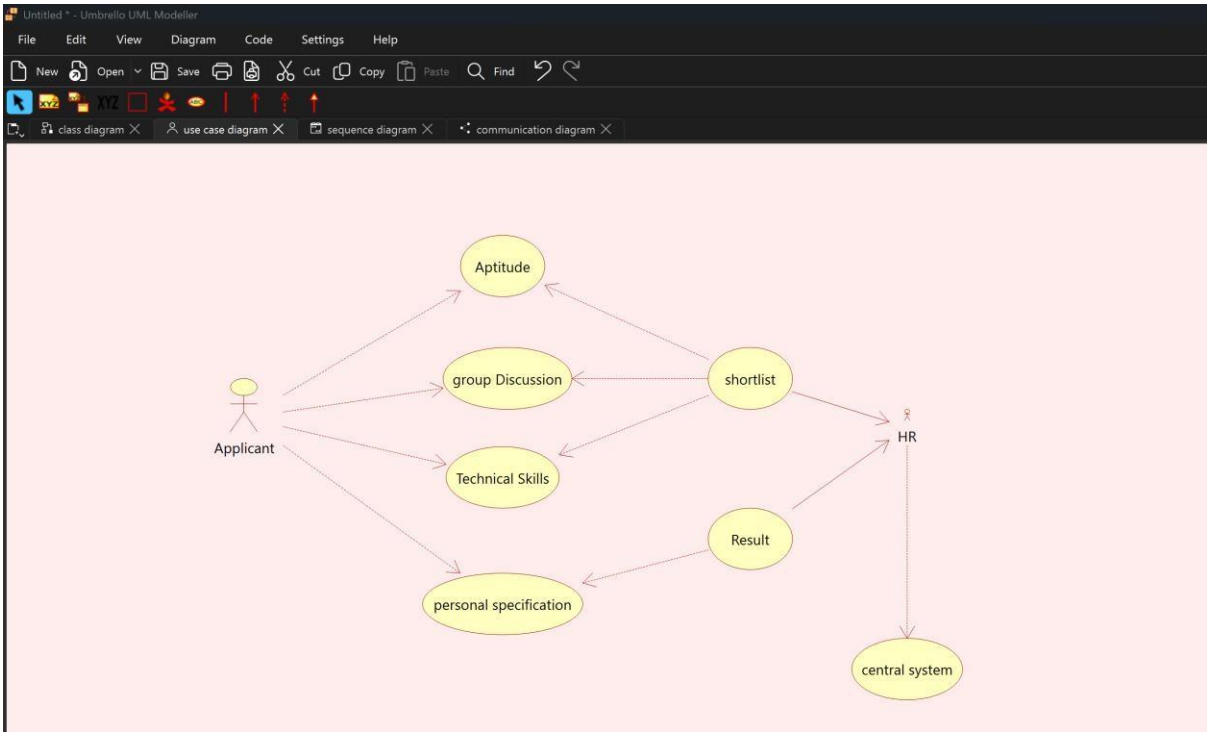
NAME: Reddy Dorniya

Reg No: 192372061

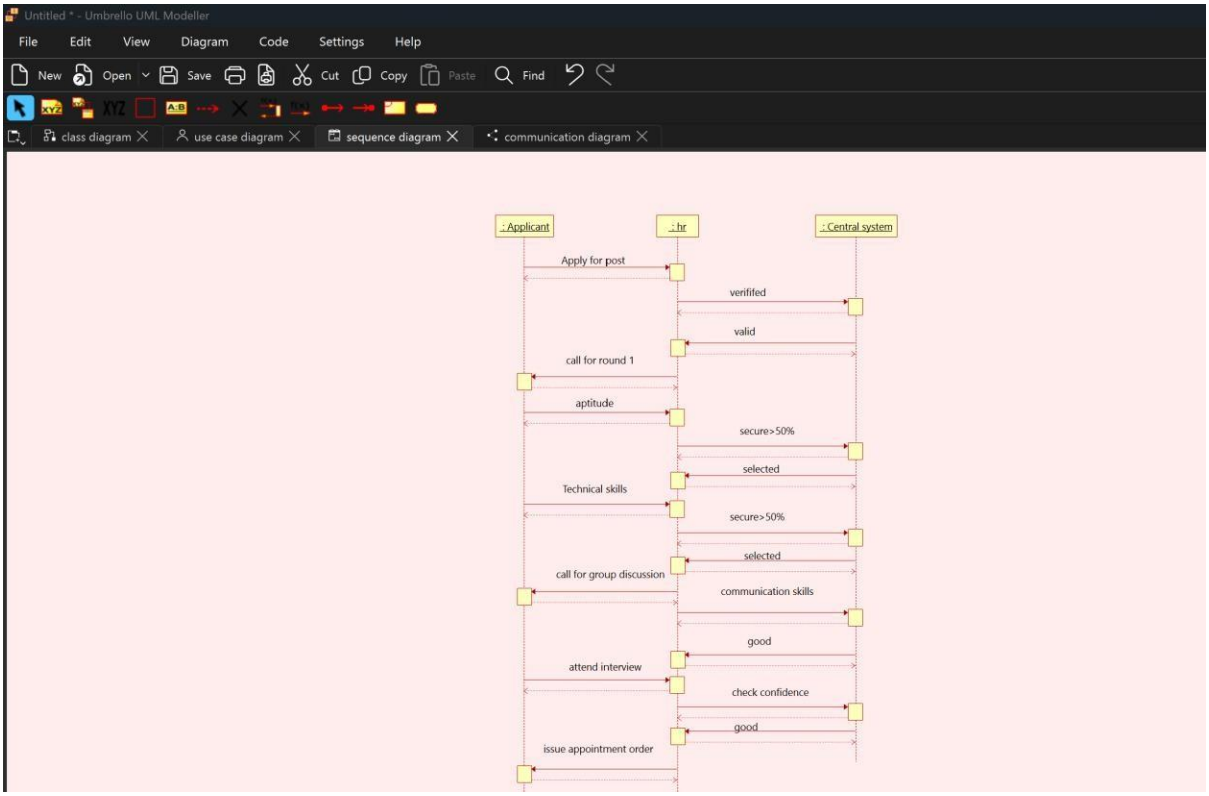
CLASS DIAGRAM:



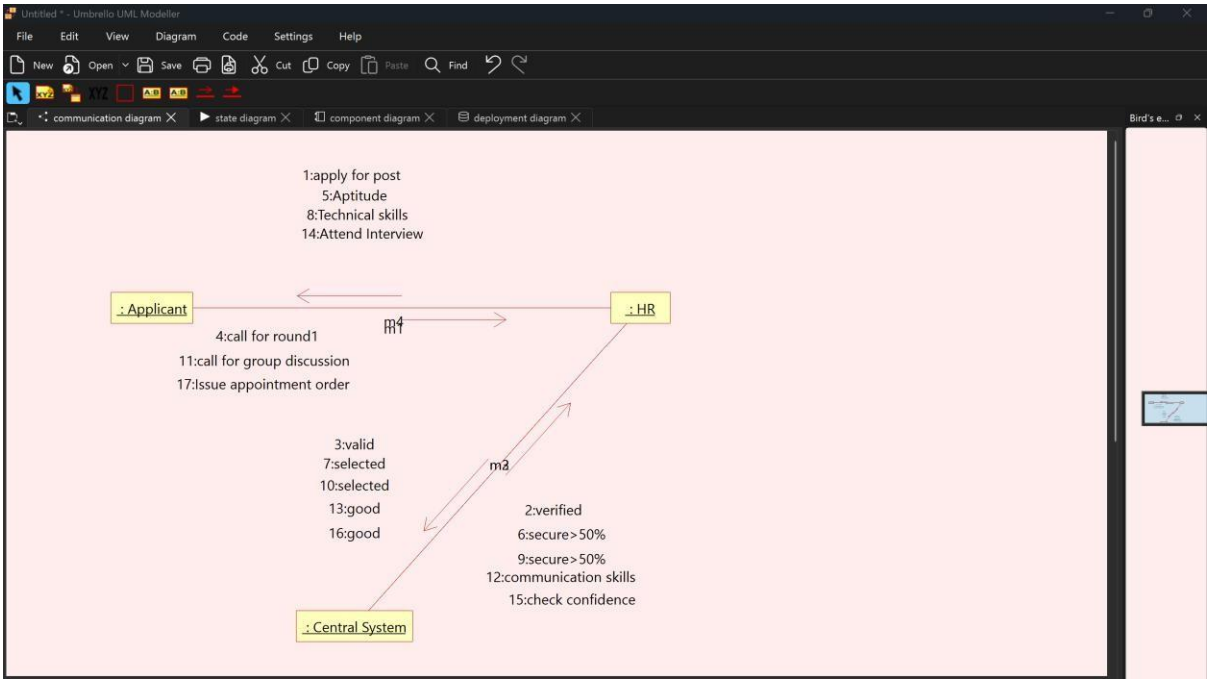
USE CASE DIAGRAM:



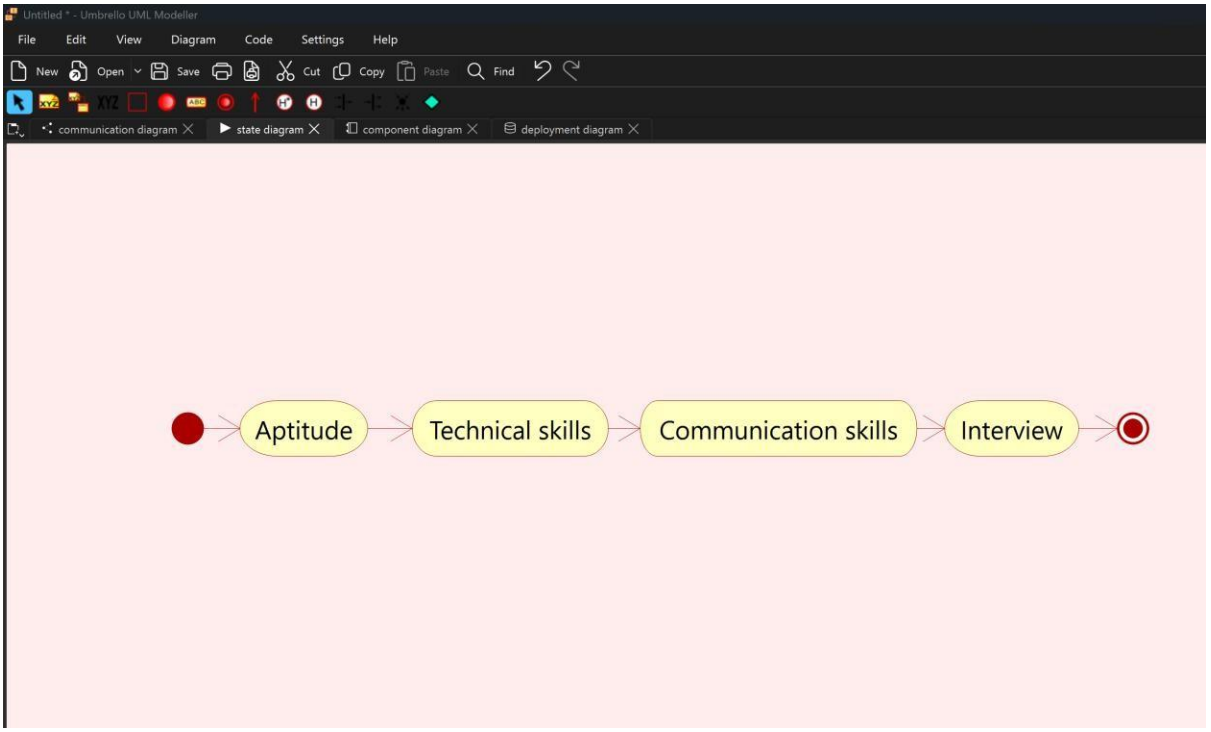
SEQUENCE DIAGRAM:



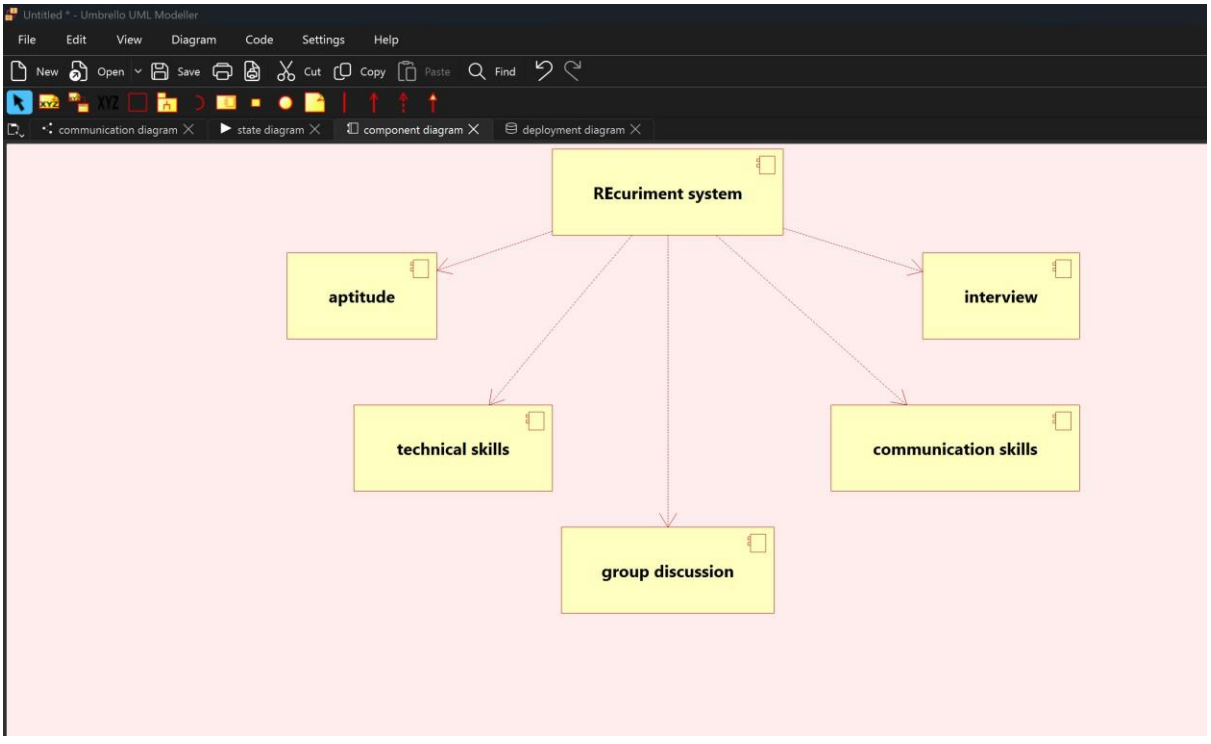
COLLABORATION DIAGRAM:



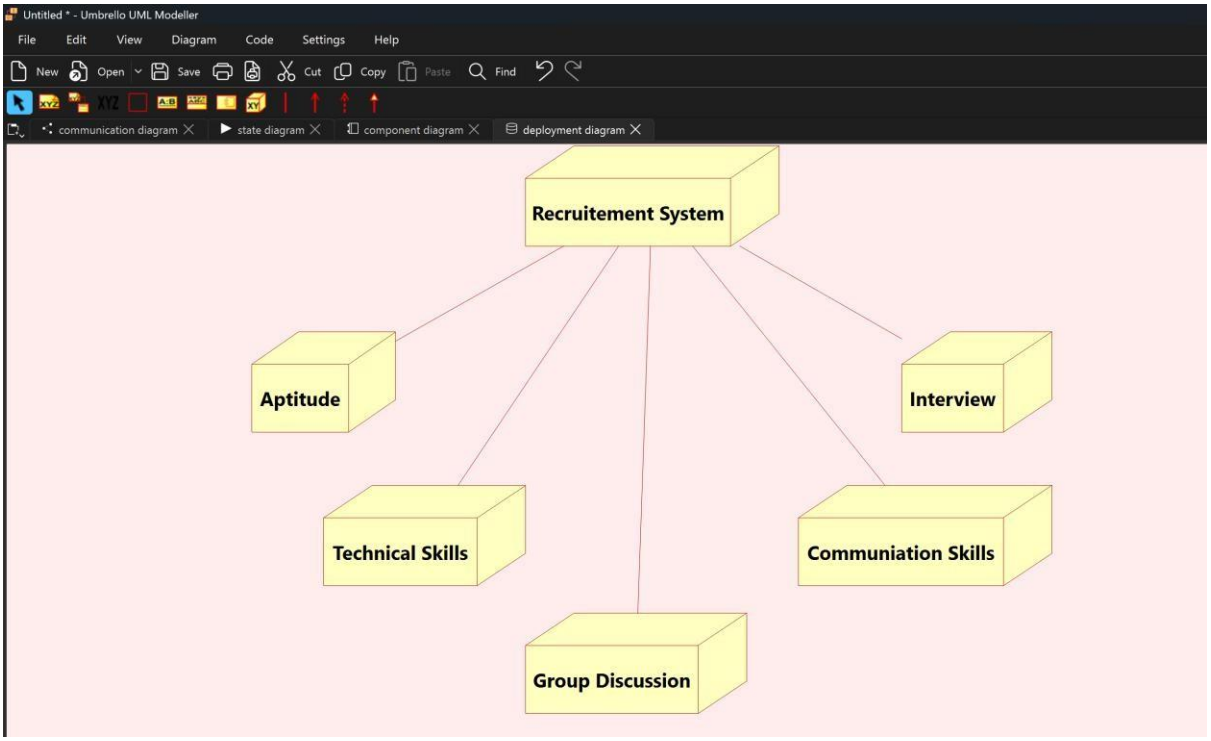
STATE DIAGRAM



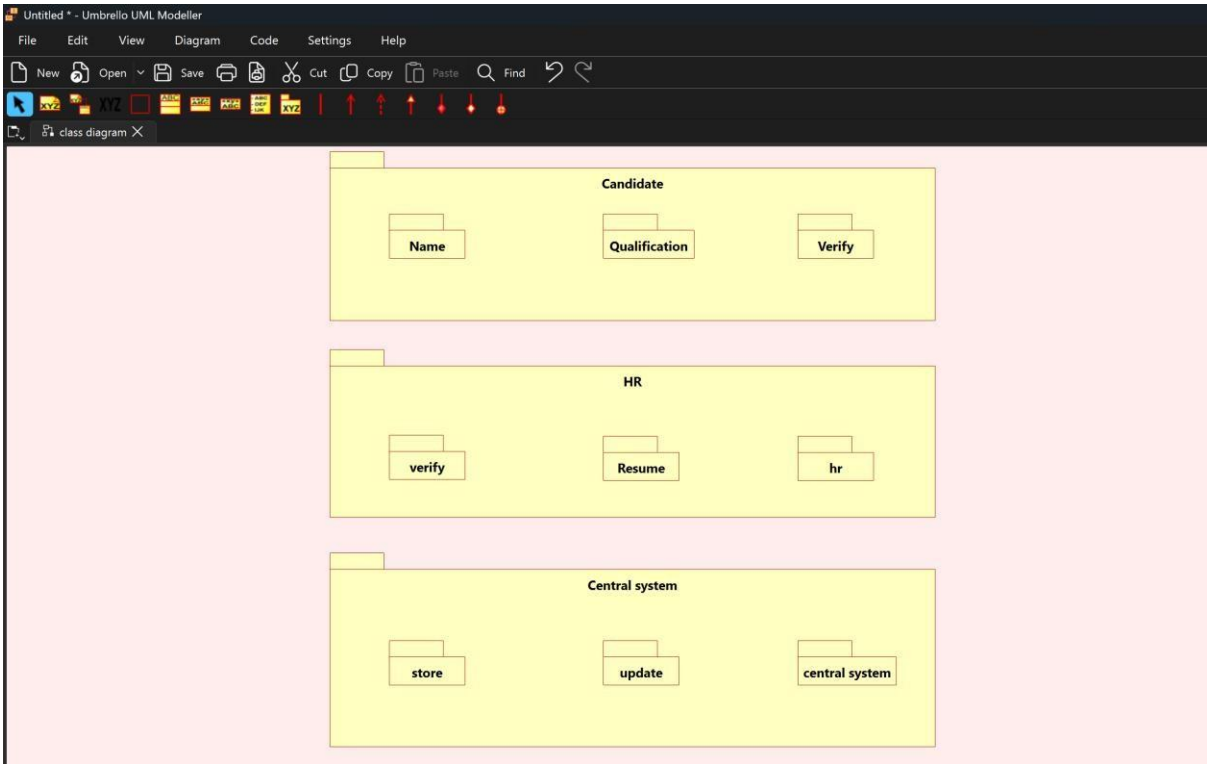
COMPONENT DIAGRAM;



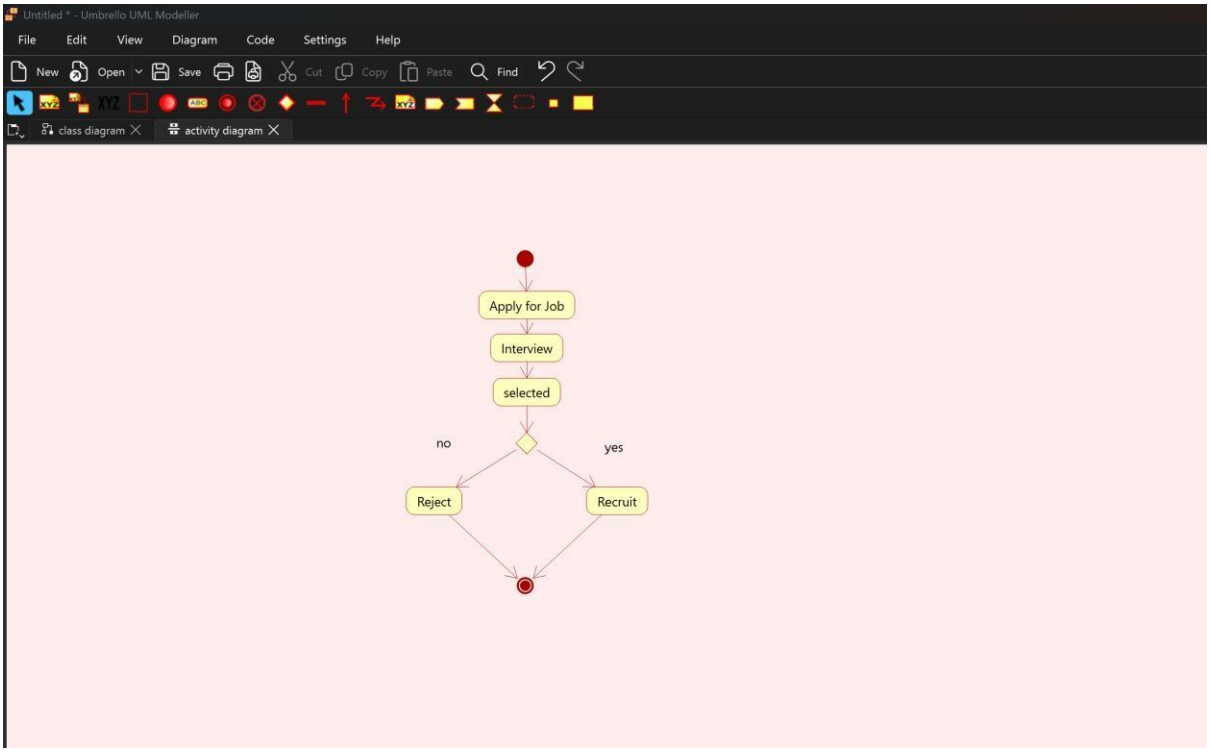
DEPLOYMENT DIAGRAM:



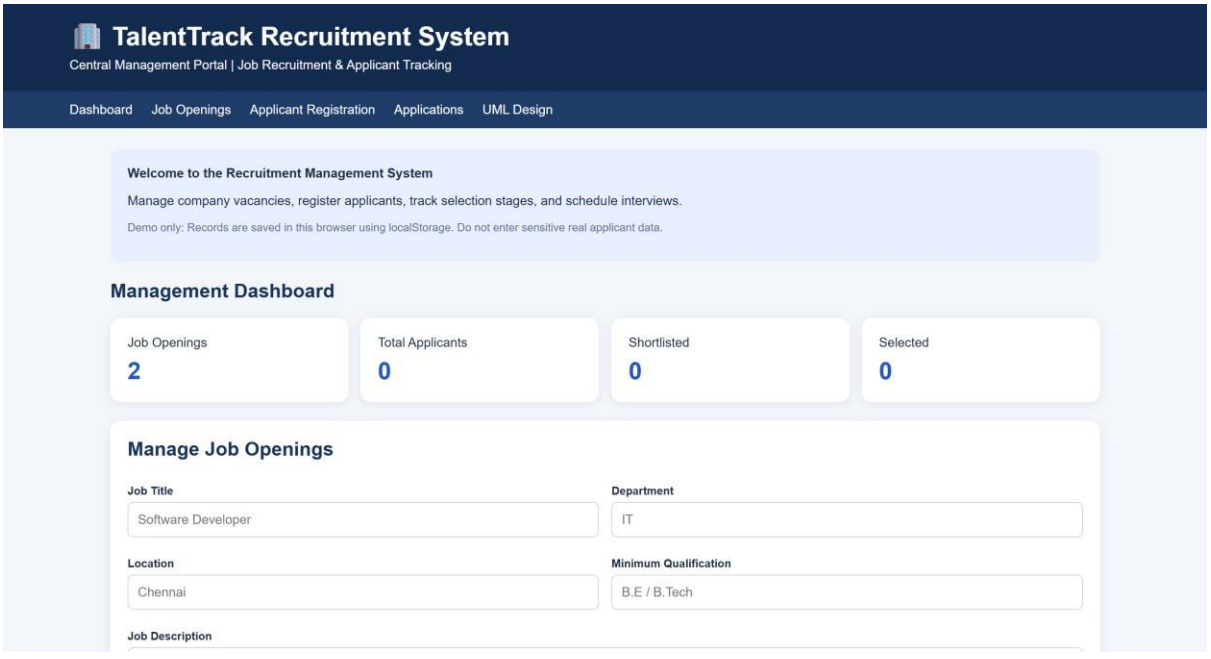
PACKAGE DIAGRAM



ACTIVITY DIAGRAM:



WEBPAGE FOR RECRUITMENT SYSTEM



Manage Job Openings

Job Title

Software Developer

Department

IT

Location

Chennai

Minimum Qualification

B.E / B.Tech

Job Description

Add Job Opening

Current Vacancies

Job ID	Title	Department	Location	Qualification	Action
JOB101	Software Developer	IT	Chennai	B.E / B.Tech	Delete
JOB102	QA Tester	Quality Assurance	Bengaluru	Any Computer Science Degree	Delete

| JOB102 | QA Tester | Quality Assurance | Bengaluru | Any Computer Science Degree | Delete |

Applicant Registration

Full Name

Email

Phone

Qualification

Years of Experience

Applying For

Select Job

Skills

Java, Python, SQL...

Submit Application

Applicant & Recruitment Management

Skills

Java, Python, SQL...

Submit Application

Applicant & Recruitment Management

Search applicant, job, or status

Application ID	Applicant	Job	Qualification	Experience	Status	Interview Date	Management Actions

UML Design – Recruitment System

Actors: Applicant, HR Manager, Central Management Administrator.

Use cases: View Jobs, Apply for Job, Manage Vacancies, Review Applicants, Shortlist Candidate, Schedule Interview, Select/Reject Candidate.

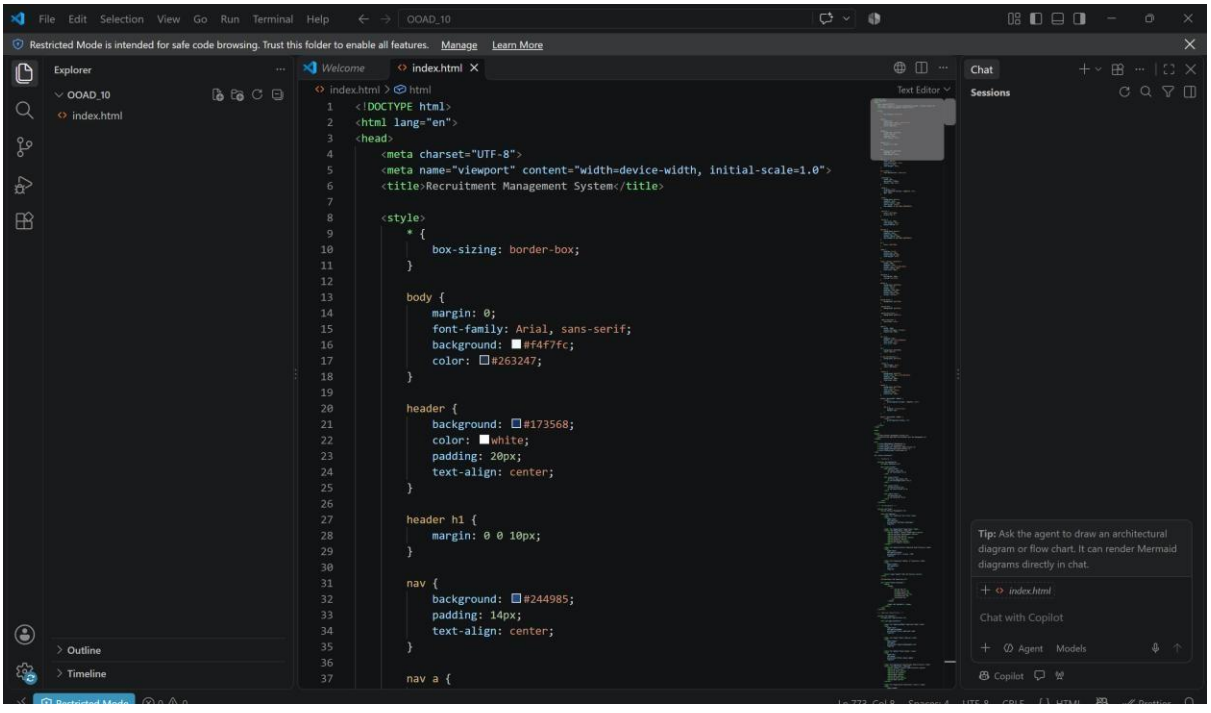
Class model: Applicant → Application → Job; HRManager manages Job, Application and Interview.

Activity flow: Post Vacancy → Applicant Registration → Application Review → Shortlist → Interview → Select or Reject.

▶ View UML use-case diagram (PlantUML source)
▶ View UML class diagram (PlantUML source)

TalentTrack | Recruitment Management System | College Project Demo

HTML CODE:




```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Recruitment Management System</title>

    <style>
        * {
            box-sizing: border-box;
        }

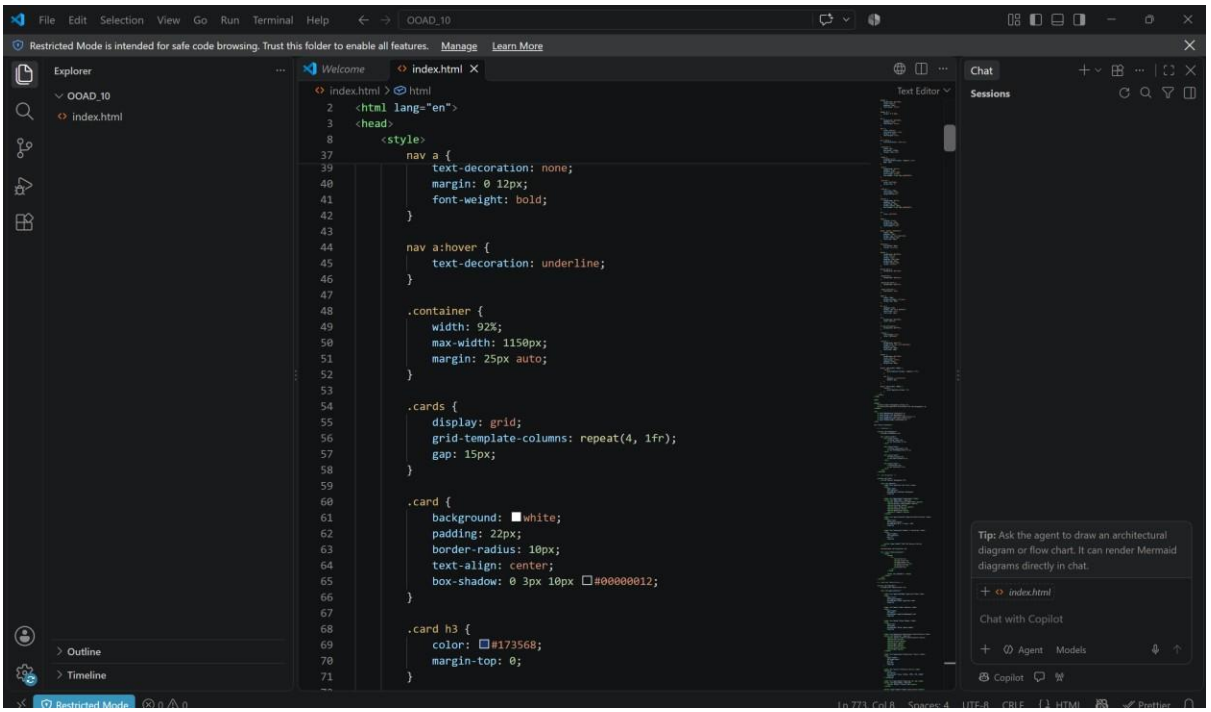
        body {
            margin: 0;
            font-family: Arial, sans-serif;
            background: #f4f7fc;
            color: #263247;
        }

        header {
            background: #173568;
            color: white;
            padding: 20px;
            text-align: center;
        }

        header h1 {
            margin: 0 0 10px;
        }

        nav {
            background: #244985;
            padding: 14px;
            text-align: center;
        }

        nav a {
```



```
<html lang="en">
<head>
    <style>
        nav a {
            text-decoration: none;
            margin: 0 12px;
            font-weight: bold;
        }

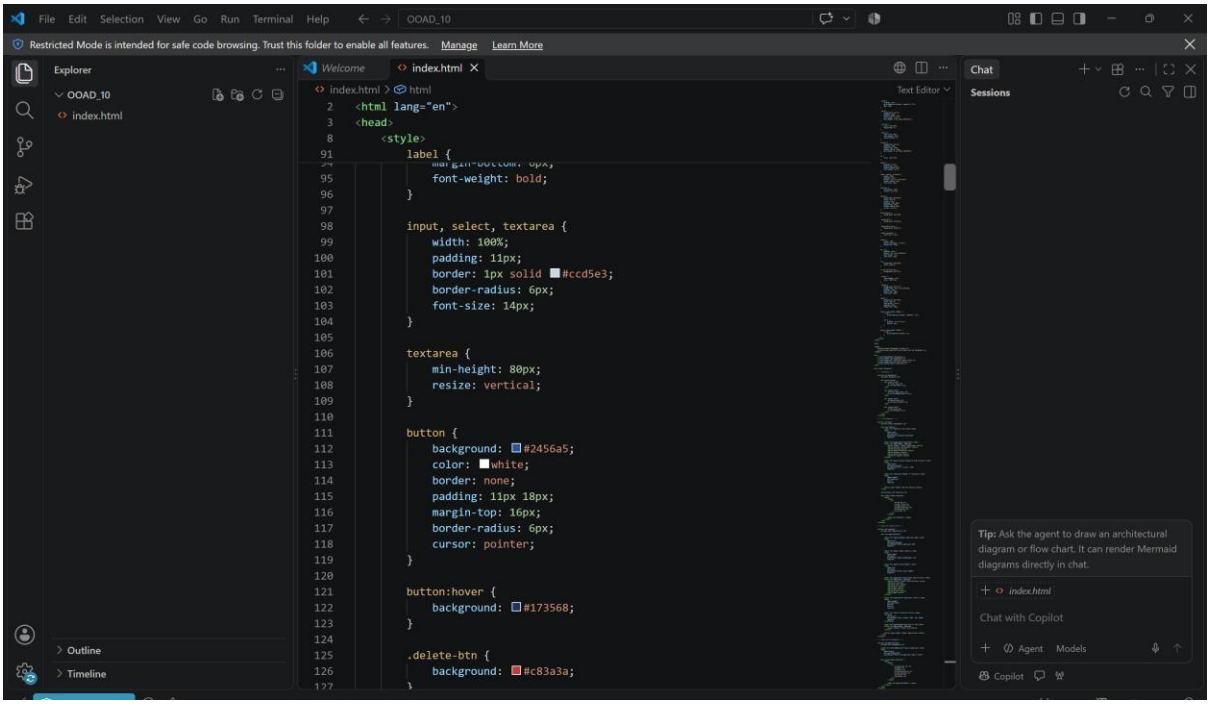
        nav a:hover {
            text-decoration: underline;
        }

        .container {
            width: 92%;
            max-width: 1150px;
            margin: 25px auto;
        }

        .cards {
            display: grid;
            grid-template-columns: repeat(4, 1fr);
            gap: 15px;
        }

        .card {
            background: white;
            padding: 22px;
            border-radius: 10px;
            text-align: center;
            box-shadow: 0 3px 10px #00000012;
        }

        .card h3 {
            color: #173568;
            margin-top: 0;
        }
```



```
<html lang="en">
<head>
    <style>
        label {
            font-weight: bold;
        }

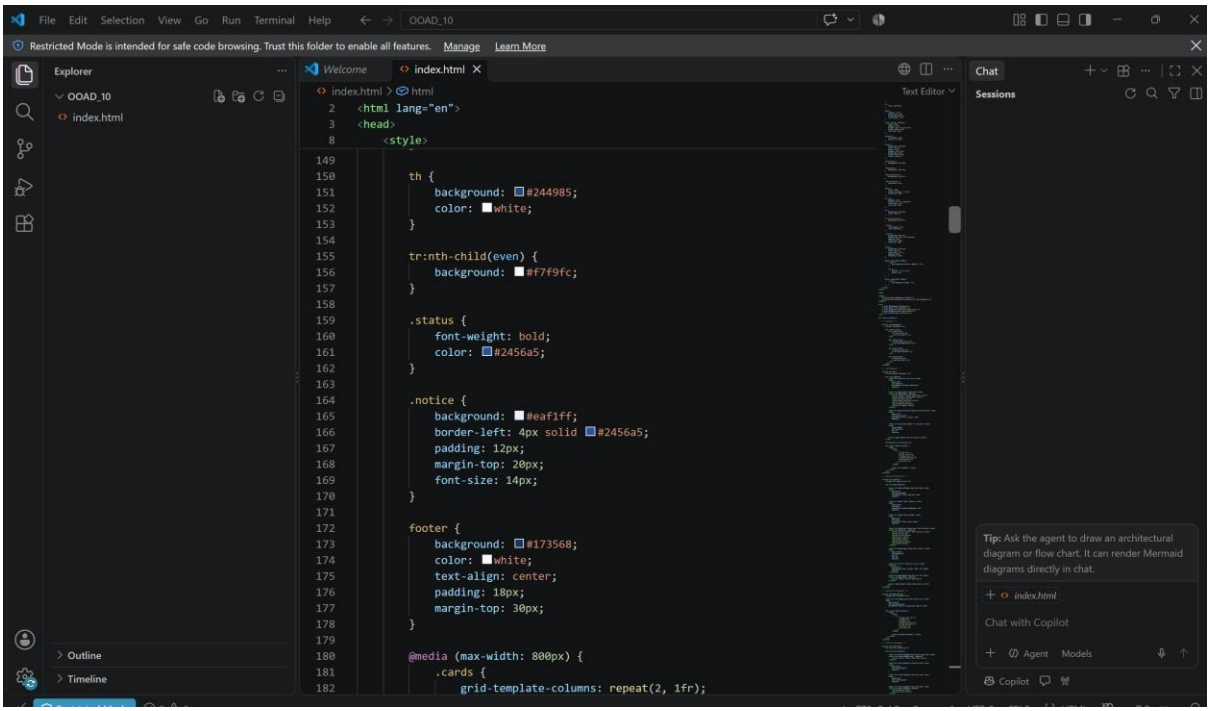
        input, select, textarea {
            width: 100%;
            padding: 11px;
            border: 1px solid #ccd5e3;
            border-radius: 6px;
            font-size: 14px;
        }

        textarea {
            min-height: 80px;
            resize: vertical;
        }

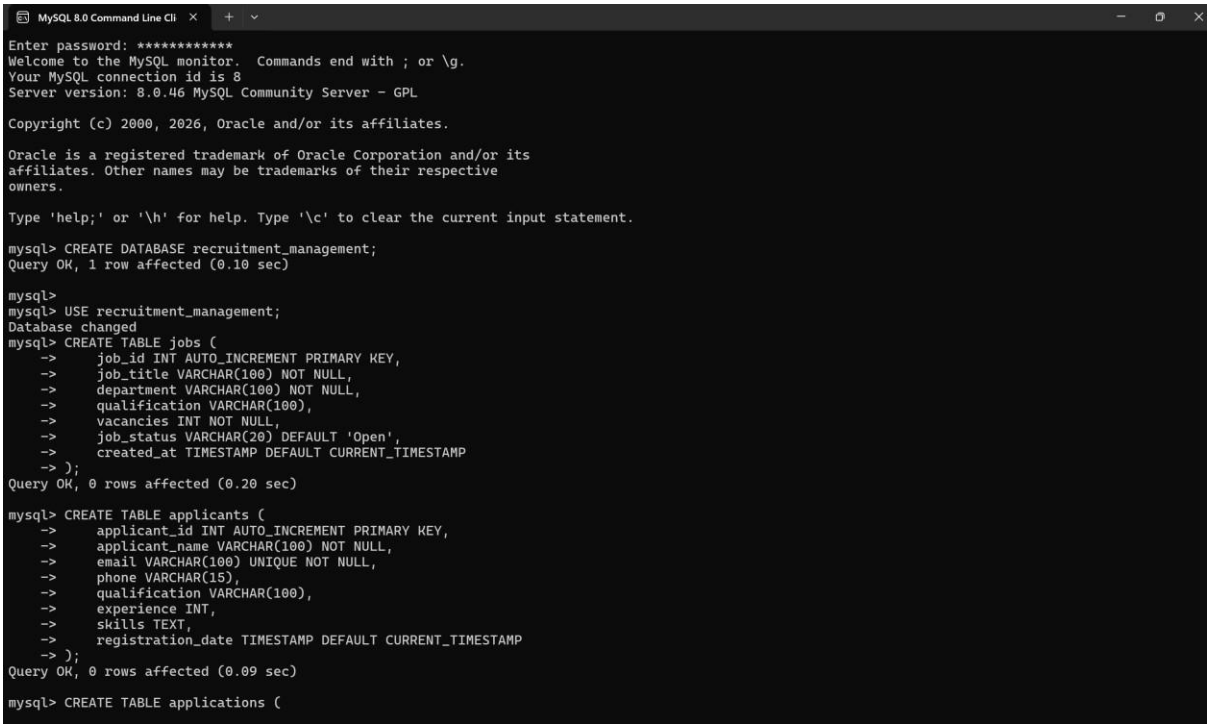
        button {
            background: #2456a5;
            color: white;
            border: none;
            padding: 11px 18px;
            margin-top: 16px;
            border-radius: 6px;
            cursor: pointer;
        }

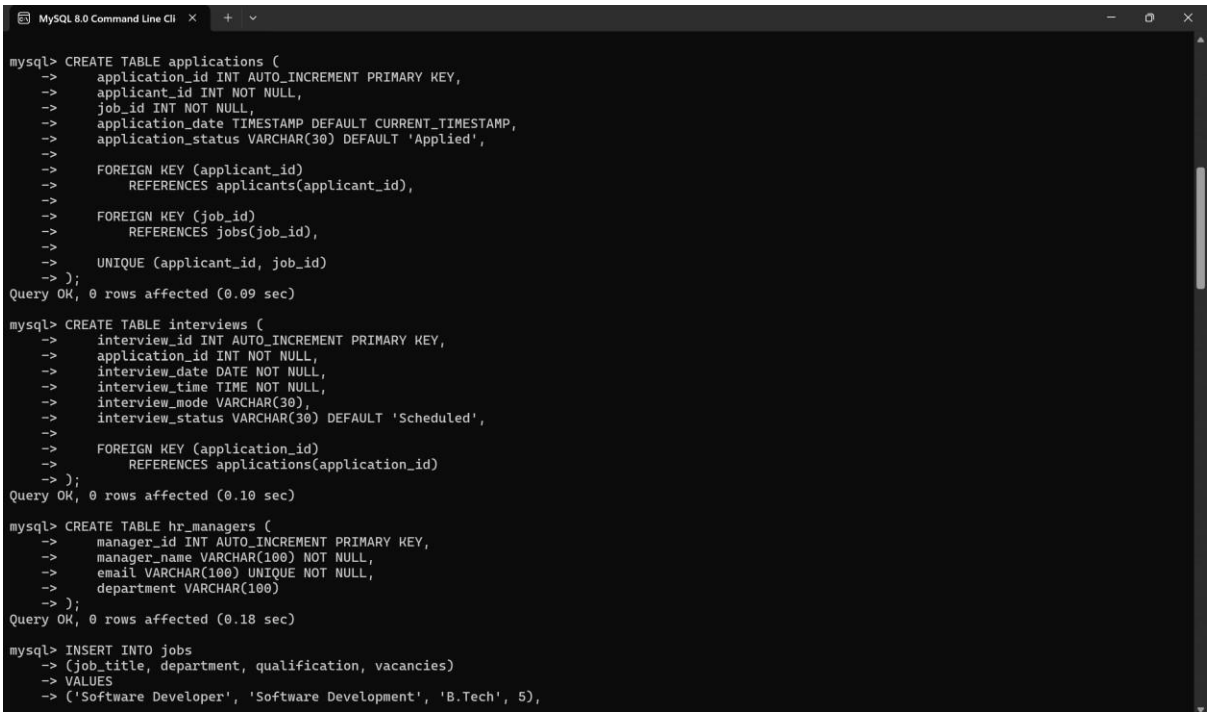
        button:hover {
            background: #173568;
        }

        .delete-btn {
            background: #c83a3a;
```



MYSQL DATABASE:



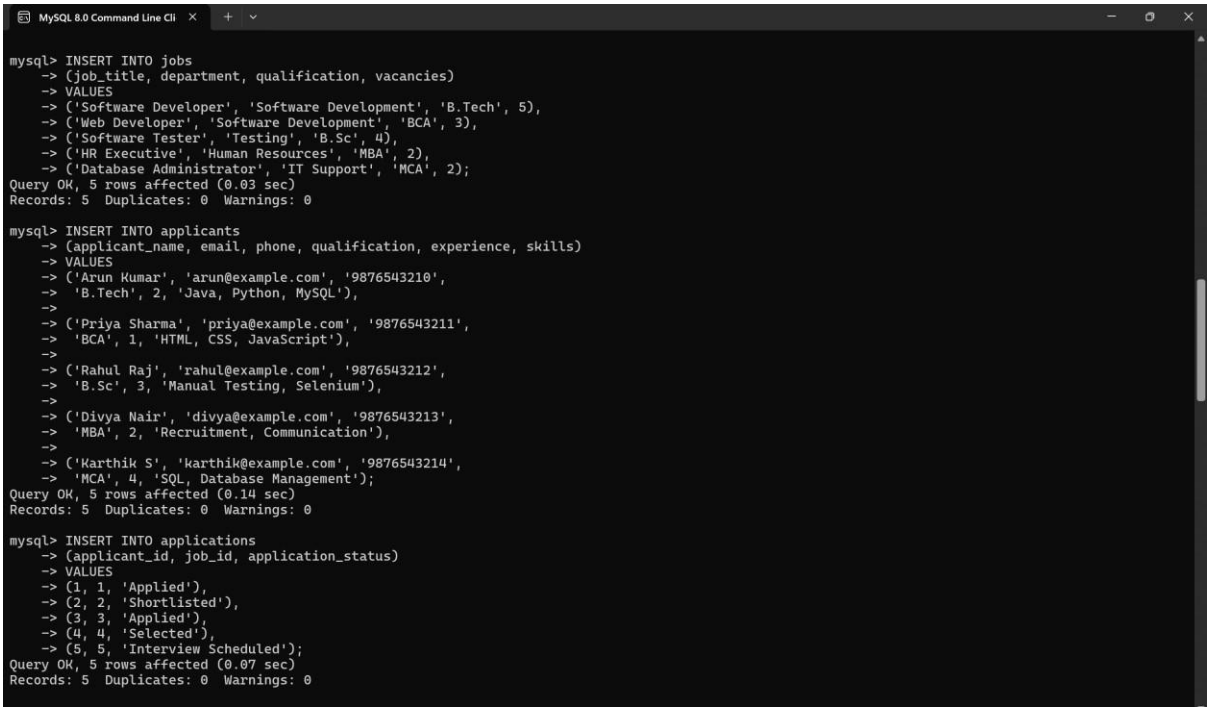



```
mysql> CREATE TABLE applications (
    ->     application_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     applicant_id INT NOT NULL,
    ->     job_id INT NOT NULL,
    ->     application_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     application_status VARCHAR(30) DEFAULT 'Applied',
    ->
    ->     FOREIGN KEY (applicant_id)
    ->         REFERENCES applicants(applicant_id),
    ->
    ->     FOREIGN KEY (job_id)
    ->         REFERENCES jobs(job_id),
    ->
    ->     UNIQUE (applicant_id, job_id)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE interviews (
    ->     interview_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     application_id INT NOT NULL,
    ->     interview_date DATE NOT NULL,
    ->     interview_time TIME NOT NULL,
    ->     interview_mode VARCHAR(30),
    ->     interview_status VARCHAR(30) DEFAULT 'Scheduled',
    ->
    ->     FOREIGN KEY (application_id)
    ->         REFERENCES applications(application_id)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE hr_managers (
    ->     manager_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     manager_name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     department VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.18 sec)

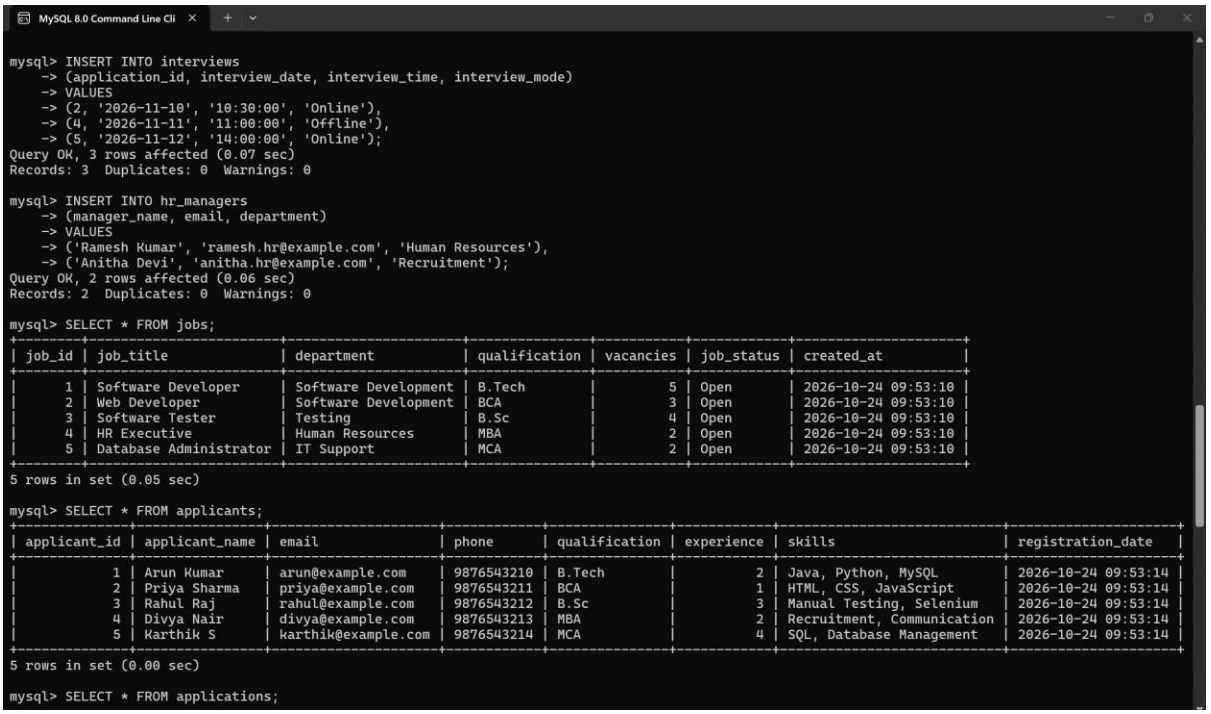
mysql> INSERT INTO jobs
    -> (job_title, department, qualification, vacancies)
    -> VALUES
    -> ('Software Developer', 'Software Development', 'B.Tech', 5),
```

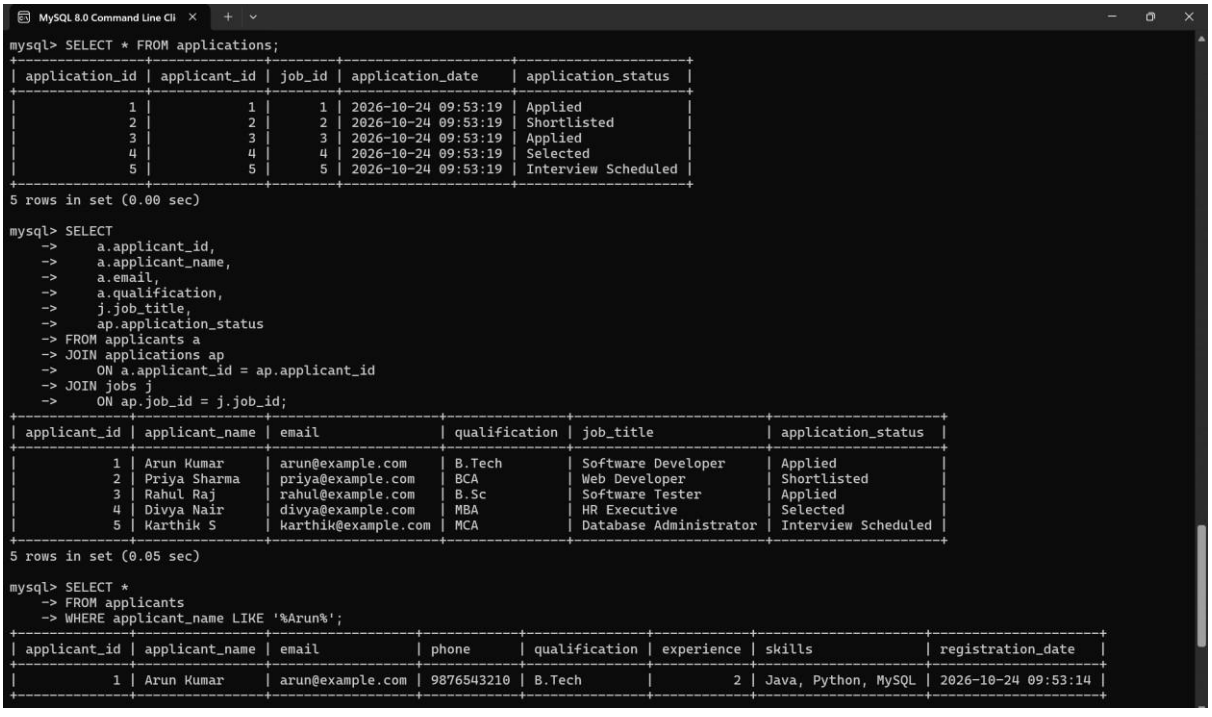


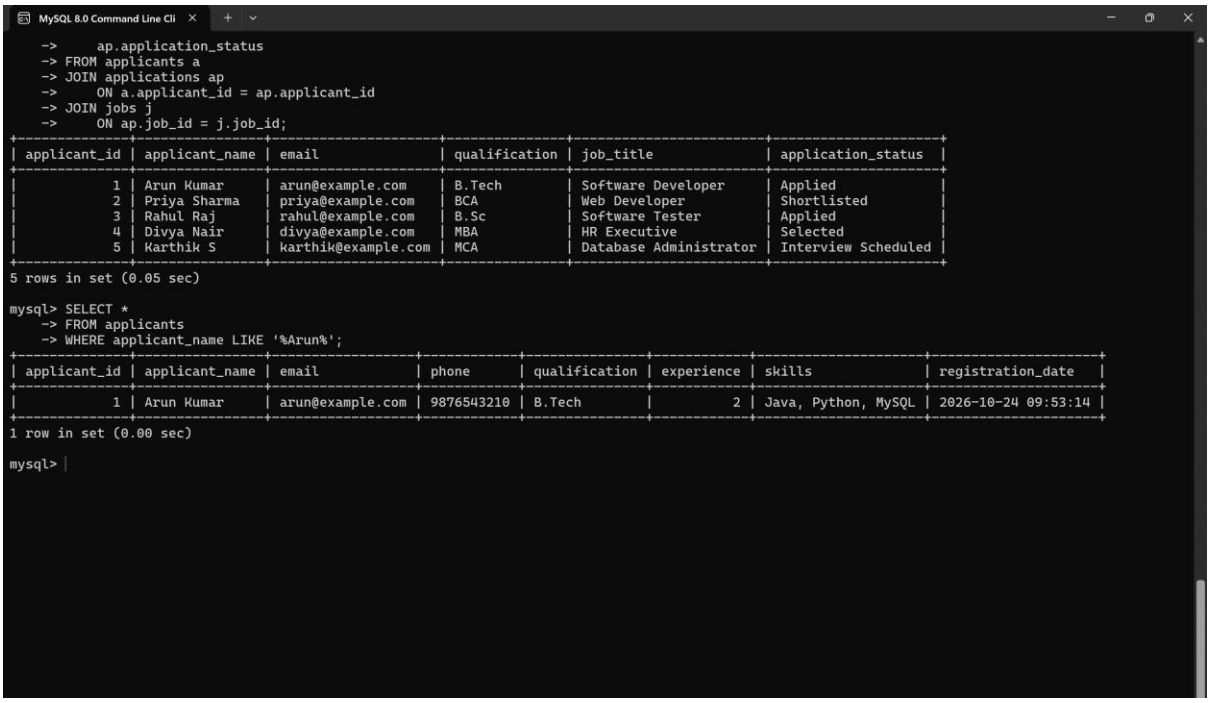

```
mysql> INSERT INTO jobs
    -> (job_title, department, qualification, vacancies)
    -> VALUES
    -> ('Software Developer', 'Software Development', 'B.Tech', 5),
    -> ('Web Developer', 'Software Development', 'BCA', 3),
    -> ('Software Tester', 'Testing', 'B.Sc', 4),
    -> ('HR Executive', 'Human Resources', 'MBA', 2),
    -> ('Database Administrator', 'IT Support', 'MCA', 2);
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO applicants
    -> (applicant_name, email, phone, qualification, experience, skills)
    -> VALUES
    -> ('Arun Kumar', 'arun@example.com', '9876543210',
    ->  'B.Tech', 2, 'Java, Python, MySQL'),
    ->
    -> ('Priya Sharma', 'priya@example.com', '9876543211',
    ->  'BCA', 1, 'HTML, CSS, JavaScript'),
    ->
    -> ('Rahul Raj', 'rahul@example.com', '9876543212',
    ->  'B.Sc', 3, 'Manual Testing, Selenium'),
    ->
    -> ('Divya Nair', 'divya@example.com', '9876543213',
    ->  'MBA', 2, 'Recruitment, Communication'),
    ->
    -> ('Karthik S', 'karthik@example.com', '9876543214',
    ->  'MCA', 4, 'SQL, Database Management');
Query OK, 5 rows affected (0.14 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO applications
    -> (applicant_id, job_id, application_status)
    -> VALUES
    -> (1, 1, 'Applied'),
    -> (2, 2, 'Shortlisted'),
    -> (3, 3, 'Applied'),
    -> (4, 4, 'Selected'),
    -> (5, 5, 'Interview Scheduled');
Query OK, 5 rows affected (0.07 sec)
Records: 5  Duplicates: 0  Warnings: 0
```


MySQL 8.0 Command Line Cli
mysql> INSERT INTO interviews
-> (application_id, interview_date, interview_time, interview_mode)
-> VALUES
-> (2, '2026-11-10', '10:30:00', 'Online'),
-> (4, '2026-11-11', '11:00:00', 'Offline'),
-> (5, '2026-11-12', '14:00:00', 'Online');
Query OK, 3 rows affected (0.07 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO hr_managers
-> (manager_name, email, department)
-> VALUES
-> ('Ramesh Kumar', 'ramesh.hr@example.com', 'Human Resources'),
-> ('Anitha Devi', 'anitha.hr@example.com', 'Recruitment');
Query OK, 2 rows affected (0.06 sec)
Records: 2 Duplicates: 0 Warnings: 0
mysql> SELECT * FROM jobs;
job_id	job_title	department	qualification	vacancies	job_status	created_at
1	Software Developer	Software Development	B.Tech	5	Open	2026-10-24 09:53:10
2	Web Developer	Software Development	BCA	3	Open	2026-10-24 09:53:10
3	Software Tester	Testing	B.Sc	4	Open	2026-10-24 09:53:10
4	HR Executive	Human Resources	MBA	2	Open	2026-10-24 09:53:10
5	Database Administrator	IT Support	MCA	2	Open	2026-10-24 09:53:10
5 rows in set (0.05 sec)
mysql> SELECT * FROM applicants;
applicant_id	applicant_name	email	phone	qualification	experience	skills	registration_date
1	Arun Kumar	arun@example.com	9876543210	B.Tech	2	Java, Python, MySQL	2026-10-24 09:53:14
2	Priya Sharma	priya@example.com	9876543211	BCA	1	HTML, CSS, JavaScript	2026-10-24 09:53:14
3	Rahul Raj	rahul@example.com	9876543212	B.Sc	3	Manual Testing, Selenium	2026-10-24 09:53:14
4	Divya Nair	divya@example.com	9876543213	MBA	2	Recruitment, Communication	2026-10-24 09:53:14
5	Karthik S	karthik@example.com	9876543214	MCA	4	SQL, Database Management	2026-10-24 09:53:14
5 rows in set (0.00 sec)
mysql> SELECT * FROM applications;


MySQL 8.0 Command Line Cli
mysql> SELECT * FROM applications;
application_id	applicant_id	job_id	application_date	application_status
1	1	1	2026-10-24 09:53:19	Applied
2	2	2	2026-10-24 09:53:19	Shortlisted
3	3	3	2026-10-24 09:53:19	Applied
4	4	4	2026-10-24 09:53:19	Selected
5	5	5	2026-10-24 09:53:19	Interview Scheduled
5 rows in set (0.00 sec)
mysql> SELECT
-> a.applicant_id,
-> a.applicant_name,
-> a.email,
-> a.qualification,
-> j.job_title,
-> ap.application_status
-> FROM applicants a
-> JOIN applications ap
-> ON a.applicant_id = ap.applicant_id
-> JOIN jobs j
-> ON ap.job_id = j.job_id;
applicant_id	applicant_name	email	qualification	job_title	application_status
1	Arun Kumar	arun@example.com	B.Tech	Software Developer	Applied
2	Priya Sharma	priya@example.com	BCA	Web Developer	Shortlisted
3	Rahul Raj	rahul@example.com	B.Sc	Software Tester	Applied
4	Divya Nair	divya@example.com	MBA	HR Executive	Selected
5	Karthik S	karthik@example.com	MCA	Database Administrator	Interview Scheduled
5 rows in set (0.05 sec)
mysql> SELECT *
-> FROM applicants
-> WHERE applicant_name LIKE '%Arun%';
| applicant_id | applicant_name | email | phone | qualification | experience | skills | registration_date |
| 1 | Arun Kumar | arun@example.com | 9876543210 | B.Tech | 2 | Java, Python, MySQL | 2026-10-24 09:53:14 |


MySQL 8.0 Command Line Cli
-> ap.application_status
-> FROM applicants a
-> JOIN applications ap
-> ON a.applicant_id = ap.applicant_id
-> JOIN jobs j
-> ON ap.job_id = j.job_id;
applicant_id	applicant_name	email	qualification	job_title	application_status
1	Arun Kumar	arun@example.com	B.Tech	Software Developer	Applied
2	Priya Sharma	priya@example.com	BCA	Web Developer	Shortlisted
3	Rahul Raj	rahul@example.com	B.Sc	Software Tester	Applied
4	Divya Nair	divya@example.com	MBA	HR Executive	Selected
5	Karthik S	karthik@example.com	MCA	Database Administrator	Interview Scheduled
5 rows in set (0.05 sec)
mysql> SELECT *
-> FROM applicants
-> WHERE applicant_name LIKE '%Arun%';
| applicant_id | applicant_name | email | phone | qualification | experience | skills | registration_date |
| 1 | Arun Kumar | arun@example.com | 9876543210 | B.Tech | 2 | Java, Python, MySQL | 2026-10-24 09:53:14 |
1 row in set (0.00 sec)
mysql>